

AI WEATHER ASSISTANT USING LANGGRAPH

Project Document

1. Introduction

AI agents can perform tasks by using tools, maintaining state, and moving through multiple steps. LangGraph is a framework for building stateful and structured agent workflows. This project demonstrates a simple AI Weather Assistant that receives a city name, obtains weather information through a weather tool, and summarizes the result for the user.

2. Objective

- To build a simple multi-step AI agent workflow.
- To understand LangGraph nodes, edges, and state.
- To demonstrate tool calling in an AI agent.
- To provide a clear weather summary to the user.

3. Technologies Used

- Python
- LangGraph
- Large Language Model (LLM)
- Weather API / Weather Tool

4. Workflow

User Input → Search/Weather Tool → Store Result in State → Summarize Result → Final Response

5. LangGraph Components

Component	Purpose
State	Stores information such as the user's city and weather result.
Nodes	Perform individual tasks such as calling the weather tool or summarizing the result.
Edges	Control the flow from one node to another.
Weather Tool	Retrieves weather information for the requested city.
LLM	Understands the request and generates a natural-language response.

6. Working of the Project

- Step 1: The user enters a city name, for example, Chennai.
- Step 2: The agent sends the request to the weather tool.
- Step 3: The weather result is stored in the workflow state.
- Step 4: The result is passed to the summarization node.
- Step 5: The LLM creates a simple weather summary.
- Step 6: The final response is shown to the user.

7. Example

Input: What is today's weather in Chennai?

Process: The agent calls the weather tool, receives the weather information, stores it in state, and

summarizes it.

Output: A concise weather report containing the available temperature and weather conditions.

8. Advantages

- Structured multi-step workflow.
- State allows information from earlier steps to be available later.
- Tools can be integrated into the agent.
- Easy to extend with additional nodes and tools.

9. Conclusion

The AI Weather Assistant demonstrates the basic concepts of LangGraph, including state, nodes, edges, and tool calling. It shows how a multi-step agent can receive a request, use an external tool, retain the result, and generate a final response.

Submitted as a project document for LangGraph / AI Agent Workflow.